LTM GSD Training

M365 Essentials

Section 1 — Fundamentals

Topic 1: Internet vs Intranet

Before understanding how Microsoft 365 works, it is important to understand two fundamental networking concepts — the Internet and the Intranet. These two terms are often confused, but they serve very different purposes.

What is the Internet?

The Internet is a massive global network that connects millions of computers, servers, and devices across the world. It is public — meaning anyone with a device and an internet connection can access it.

The Internet allows people to browse websites, send emails, stream videos, access cloud services, and communicate across the globe. There is no single owner of the Internet — it is maintained collectively by internet service providers (ISPs), governments, and organizations worldwide.

Key characteristics of the Internet:

- Publicly accessible by anyone around the world.
- Connects billions of devices across different countries and networks.
- Uses standard protocols such as TCP/IP, HTTP, HTTPS, and DNS.
- Always on and globally available — 24 hours a day, 7 days a week.
- Examples of internet-based services: Google, YouTube, Microsoft 365, Gmail, social media.

What is an Intranet?

An Intranet is a private network that exists within an organization. It uses the same technology as the Internet but is accessible only to authorized employees or members of that organization. It is not open to the public.

Organizations use intranets to share internal information, documents, policies, and tools securely — without exposing that information to the outside world.

Key characteristics of an Intranet:

- Private — accessible only to authorized users within the organization.
- Used to share internal resources such as HR policies, company announcements, shared drives, and internal tools.
- Secured by firewalls and access controls that prevent unauthorized external access.
- Can be accessed from within the office or remotely through a VPN.
- Example: A company's internal HR portal where employees can access payslips, apply for leave, or read company policies.

> **Analogy:** Think of the Internet as a public highway — anyone can drive on it. An Intranet is like a private road inside a gated community — only residents (employees) have access, and it is protected from outsiders.

Internet vs Intranet — Side by Side

Feature	Internet vs Intranet
Access	Internet: Open to everyone globally. Intranet: Restricted to authorized users only.
Ownership	Internet: No single owner — maintained globally. Intranet: Owned and managed by the organization.
Purpose	Internet: General information, communication, and cloud services. Intranet: Internal collaboration, documents, and tools.
Security	Internet: Public — requires extra security measures. Intranet: Private — protected by firewalls and access controls.
Examples	Internet: Google, Microsoft 365, YouTube. Intranet: Company HR portal, internal knowledge base, SharePoint sites.

Why This Matters for M365: Microsoft 365 is delivered over the Internet — employees access their emails, files, and collaboration tools through the internet. However, organizations often use SharePoint as their Intranet — a private internal website that only employees can access. Understanding this distinction helps you support users more effectively.

Topic 2: What is a Data Center?

When you send an email, save a file to OneDrive, or join a Teams meeting — where does that data actually go? It goes to a Data Center. Understanding what a data center is helps you understand the foundation on which cloud services like Microsoft 365 are built.

Definition

A Data Center is a large, highly secure physical facility that houses hundreds or thousands of servers, storage systems, networking equipment, and supporting infrastructure. These servers store, process, and deliver data and applications to users over the internet.

In simple terms — a data center is a giant room (or building) full of powerful computers that run the applications and store the data that organizations and individuals rely on every day.

Analogy: Think of a data center like a massive, secure warehouse for digital information. Instead of storing physical goods like boxes and crates, it stores data, applications, and services. And just like a warehouse has forklifts, loading bays, and security guards, a data center has servers, network cables, power systems, and 24/7 security.

What is Inside a Data Center?

A modern data center contains several critical components that work together to keep services running reliably:

Component	Purpose
Servers	Powerful computers that store and process data. Thousands of servers run simultaneously inside a large data center.
Storage Systems	High-capacity disk arrays and SSDs that store vast amounts of data reliably and with fast access speeds.
Networking Equipment	Routers, switches, and cables that connect servers to each other and to the internet, ensuring fast data transfer.
Power Supply (UPS)	Uninterruptible Power Supply systems ensure the data center never loses power. Backup generators kick in if the main power fails.
Cooling Systems	Servers generate enormous heat. Cooling systems (air conditioning, liquid cooling) keep temperatures at safe operating levels.
Physical Security	Data centers have strict access controls — biometric scanners, CCTV cameras, security guards, and restricted entry zones.
Fire Suppression Systems	Specialized fire suppression systems protect equipment without using water, which would damage hardware.

Types of Data Centers

Not all data centers are the same. They can be categorized based on who owns and operates them:

Type	Description	Example
Enterprise / Private Data Center	Built and owned by a single organization for its own use.	A bank running its own data center to store customer records.
Colocation Data Center	A shared facility where multiple organizations rent space to place their own servers.	A company renting server racks in a shared data center building.

Type	Description	Example
Cloud / Hyperscale Data Center	Massive facilities owned by cloud providers, serving millions of customers worldwide.	Microsoft Azure, Amazon AWS, Google Cloud data centers.
Edge Data Center	Smaller data centers placed closer to end users to reduce latency for time-sensitive services.	A small data center in a city to serve local streaming or IoT services.

Microsoft's Global Data Centers

Microsoft operates one of the largest networks of data centers in the world to power its cloud services — including Microsoft 365, Azure, and Teams.

- Microsoft has data centers in over 60 regions across the globe.
- In India, Microsoft operates data centers in Pune, Mumbai, and Chennai.
- These data centers ensure that your Microsoft 365 data is stored, backed up, and available with a guaranteed uptime of 99.9%.
- Data stored in Indian data centers stays within India, which helps organizations meet local data residency and compliance requirements.

> **Key Point:** When you use Microsoft 365, you are not storing data on your office computer or local server. You are storing it on Microsoft's data centers. This is exactly what 'the cloud' means — your data lives on someone else's highly secure, always-on servers located in these massive facilities.

Topic 3: Cloud Computing

Now that you understand what a data center is, let us understand what Cloud Computing actually means — because Microsoft 365 is entirely built on this concept.

What is Cloud Computing?

Cloud Computing is the delivery of computing services — such as servers, storage, databases, networking, software, and intelligence — over the Internet, on demand. Instead of buying and maintaining your own physical hardware and software, you access these resources from a cloud provider's data centers through the internet and pay only for what you use.

> **Simple Definition:** Cloud Computing means using someone else's computers and infrastructure (over the internet) to run your applications, store your data, and deliver your services — instead of doing it all yourself.

Before cloud computing existed, organizations had to:

- Buy expensive servers and set them up in their own offices or data centers.
- Hire IT staff to install, configure, and maintain all the hardware and software.
- Plan years ahead for capacity — if more users joined, they had to buy more servers in advance.
- Manage power, cooling, security, and physical space for all equipment.

Cloud computing eliminated most of these challenges by moving all of this to a provider like Microsoft, Amazon, or Google.

Key Characteristics of Cloud Computing

The National Institute of Standards and Technology (NIST) defines cloud computing through five essential characteristics. Understanding these helps you explain what makes something truly 'cloud':

Characteristic	What It Means
On-Demand Self-Service	Users can provision and access computing resources (storage, servers, apps) whenever they need, without requiring human interaction with the provider. Example: An admin can add 50 new Microsoft 365 user licenses instantly from the admin portal.
Broad Network Access	Services are available over the internet and accessible from any device — laptops, phones, tablets, or desktops — from anywhere in the world.

Characteristic	What It Means
Resource Pooling	The cloud provider's resources (servers, storage) are shared across many customers simultaneously, but each customer's data remains separate and secure. Multiple organizations use the same underlying infrastructure.
Rapid Elasticity	Resources can be scaled up or down quickly based on demand. Organizations can add more users or storage in minutes, and reduce them just as quickly. You only pay for what you actually use.
Measured Service	Cloud usage is monitored, measured, and reported transparently. Organizations are billed based on actual consumption — like a utility bill for electricity or water.

Why Organizations Move to the Cloud

The shift from traditional IT to cloud computing is driven by clear practical benefits:

Benefit	Explanation
Cost Savings	No upfront hardware costs. Pay monthly or yearly only for what you use. No need to invest in expensive server infrastructure.
Always Up to Date	The cloud provider handles all updates, patches, and upgrades. Users always have the latest version without any manual intervention.
Scalability	Start small and grow as needed. Adding 100 new users to Microsoft 365 takes minutes — not weeks of procurement and setup.
High Availability	Cloud providers guarantee high uptime (Microsoft 365 guarantees 99.9%). Built-in redundancy means services rarely go down.
Disaster Recovery	Data is automatically replicated across multiple data centers. If one data center fails, services continue from another.
Remote Work Support	Employees can access all their tools and data from anywhere in the world, on any device — essential for modern hybrid work environments.
Security	Cloud providers invest billions in security. Microsoft 365 includes built-in threat protection, encryption, and compliance tools.

> **Important Distinction:** Moving to the cloud does not mean your data is 'floating in the air'. It is physically stored on servers inside Microsoft's data centers. The word 'cloud' is simply a metaphor for accessing computing resources over the internet.

Topic 4: Cloud Service Models — IaaS, PaaS and SaaS

Cloud computing is not a single thing — it comes in different forms depending on how much of the technology stack the provider manages versus how much the customer manages. These are called Cloud Service Models.

There are three main service models: IaaS, PaaS, and SaaS. Each one represents a different level of control and responsibility. Understanding these models is essential because Microsoft 365 falls squarely into one of them.

> **Helpful Analogy — Think of Pizza:** Imagine you want to eat pizza. You have three options: (1) Make it completely yourself at home — buy ingredients, make the dough, bake it. (2) Buy a ready-made frozen pizza — someone else made it, you just bake it. (3) Order from a restaurant — they do everything, you just eat it. IaaS is option 1, PaaS is option 2, and SaaS is option 3.

IaaS — Infrastructure as a Service

In IaaS, the cloud provider gives you the raw computing infrastructure — virtual machines (servers), storage, and networking. You are responsible for everything that runs on top of it — the operating system, middleware, applications, and data.

What the provider manages:

- Physical servers, data centers, networking hardware, and virtualization.

What you (the customer) manage:

- Operating system (Windows, Linux).
- Middleware and runtime environments.
- Applications and data.
- Security configurations on the OS and applications.

Best suited for:

- Organizations that want the flexibility of building their own environment without buying physical hardware.
- Development and testing environments.
- Organizations migrating from on-premises infrastructure to the cloud.

> **Example:** Microsoft Azure Virtual Machines is an IaaS product. You rent a virtual server from Microsoft, install Windows Server on it, configure it the way you want, and run your own applications on it.

PaaS — Platform as a Service

In PaaS, the cloud provider manages the infrastructure AND the underlying platform (operating system, runtime, middleware). You only manage your application and data. This lets developers focus purely on building applications without worrying about infrastructure.

What the provider manages:

- Physical servers, networking, virtualization, operating system, middleware, and runtime.

What you (the customer) manage:

- Your application and its data.

Best suited for:

- Developers who want to build and deploy applications quickly without managing servers.
- Organizations building custom web applications or APIs.

> **Example:** Microsoft Azure App Service is a PaaS product. A developer writes a web application and deploys it to Azure — Microsoft handles all the servers, OS, and patching automatically.

SaaS — Software as a Service

In SaaS, the cloud provider manages everything — the infrastructure, platform, and the application itself. The customer simply uses the software through a web browser or app. There is nothing to install or maintain.

What the provider manages:

- Everything — physical servers, networking, OS, middleware, application, and updates.

What you (the customer) manage:

- Your data and user access settings.
- Configuration of the application to meet your organization's needs.

Best suited for:

- Organizations that want to use ready-made software without the overhead of managing it.
- Email, collaboration, productivity, and communication tools.

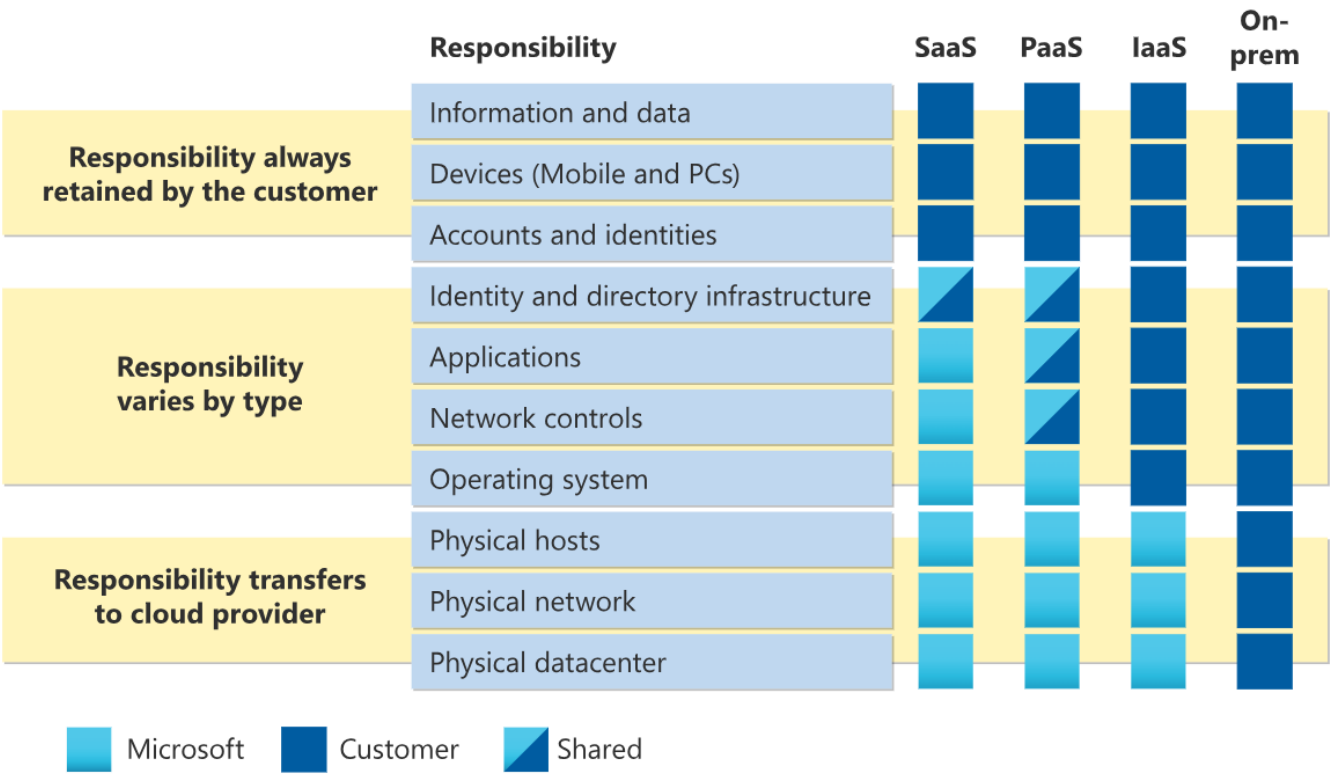
> **Example:** Microsoft 365, Gmail, Salesforce, and Zoom are all SaaS products. You open a browser, sign in, and start using the application. Microsoft manages everything behind the scenes.

Comparing IaaS, PaaS, and SaaS

Aspect	IaaS	PaaS	SaaS
What you get	Virtual machines, storage, networking	Development platform and runtime	Ready-to-use software application
You manage	OS, apps, data, security configs	Application and data only	Data and user settings only

Aspect	IaaS	PaaS	SaaS
Provider manages	Physical hardware and virtualization	Hardware, OS, middleware, and runtime	Everything — hardware to application
Control level	Highest — most flexible	Medium — focus on app development	Lowest — just use the software
Technical skill needed	High — requires IT expertise	Medium — developer knowledge needed	Low — any user can operate it
Example	Azure Virtual Machines, AWS EC2	Azure App Service, Google App Engine	Microsoft 365, Gmail, Salesforce

Where Microsoft 365 Fits: Microsoft 365 is a SaaS product. Microsoft manages the entire infrastructure, platform, and applications. As an organization, you simply assign licenses to users and they start using Outlook, Teams, SharePoint, and other apps. You never install or maintain a single server for these services.



Topic 5: Cloud Delivery Models — Public, Private and Hybrid

While service models (IaaS, PaaS, SaaS) define what type of service is provided, delivery models define how and where the cloud infrastructure is deployed. There are three main delivery models: Public Cloud, Private Cloud, and Hybrid Cloud.

Public Cloud

In a Public Cloud, the cloud infrastructure is owned and operated by a third-party provider (like Microsoft, Amazon, or Google) and shared across multiple organizations over the internet. The provider manages all the hardware and infrastructure, and customers access services through the internet.

Key characteristics:

- Infrastructure is owned and maintained entirely by the cloud provider.
- Multiple organizations (called 'tenants') share the same underlying infrastructure, but their data remains separate and isolated.

- Customers pay only for what they use — no capital investment required.
- Highly scalable — resources can be added or removed instantly.
- Accessible from anywhere with an internet connection.

Advantages:

- No upfront hardware costs — fully pay-as-you-go.
- Zero maintenance — the provider handles all updates and repairs.
- Massive scalability — can support thousands of users instantly.
- High reliability — built-in redundancy and disaster recovery.

Considerations:

- Data is stored on shared infrastructure outside the organization's direct control.
- Organizations with strict data sovereignty requirements may need to verify data residency options.

> **Example:** Microsoft 365 and Microsoft Azure are Public Cloud services. Your organization uses Microsoft's shared data centers. Your data is isolated and secure, but the physical hardware is shared with other Microsoft customers worldwide.

Private Cloud

In a Private Cloud, the cloud infrastructure is dedicated exclusively to a single organization. It can be hosted on the organization's own premises (on-site) or in a dedicated section of a third-party data center — but it is not shared with anyone else.

Key characteristics:

- Infrastructure is used by only one organization — not shared.
- Can be managed internally by the organization's IT team or by a managed service provider.
- Offers maximum control and customization over the environment.
- Higher cost compared to public cloud — the organization bears the full cost of hardware and maintenance.

Best suited for:

- Government agencies and defense organizations with strict data sovereignty requirements.
- Banks and financial institutions that must keep data within their own infrastructure for regulatory reasons.
- Healthcare organizations handling sensitive patient data that must meet strict compliance standards.

> **Example:** A large bank builds and maintains its own cloud infrastructure within its own data center. All cloud-like benefits (virtualization, scalability, self-service provisioning) are available, but the hardware belongs entirely to the bank.

Hybrid Cloud

A Hybrid Cloud combines both Public and Private Cloud environments, connected and working together. Organizations use a private cloud (or on-premises infrastructure) for sensitive, critical workloads and the public cloud for everything else.

Key characteristics:

- A mix of on-premises (or private cloud) and public cloud resources.
- Both environments are connected — data and applications can move between them.
- Organizations retain control over sensitive data while still benefiting from the scalability of the public cloud.
- Ideal for organizations in the middle of a cloud migration journey.

Best suited for:

- Organizations that have legacy systems on-premises they cannot immediately migrate to the cloud.
- Businesses that need cloud scalability for some workloads but must keep certain data on-premises for compliance.
- Organizations gradually transitioning from fully on-premises to fully cloud over time.

> **Example:** A hospital runs its patient records management system on a private on-premises server (due to compliance), but uses Microsoft 365 (public cloud) for email, Teams, and collaboration. This combined setup is a Hybrid Cloud environment.

Topic 6: On-Premises vs Cloud Infrastructure

Now that you understand what cloud computing is and its different models, it is important to compare it directly with the traditional way organizations managed their IT — called On-Premises infrastructure. This comparison will help you clearly understand why organizations are moving to the cloud and what changes when they do.

What is On-Premises Infrastructure?

On-Premises (often abbreviated as On-Prem) refers to the traditional model where an organization buys, installs, and manages all its own IT hardware and software within its own physical premises — usually in a server room or a private data center within the office building.

In an on-premises environment:

- The organization owns all the servers, storage, and networking hardware.
- The IT team is responsible for setting up, maintaining, patching, and upgrading everything.
- Applications (like email servers, file servers, and directory services) are installed and run locally.
- All data is stored on hardware that the organization physically controls.
- Users connect to systems over the internal network (LAN) or via VPN when working remotely.

> **Example:** Before Microsoft 365 existed, organizations ran Microsoft Exchange Server on their own physical servers inside the office to manage emails. The IT team was responsible for maintaining that server, applying patches, managing storage, and ensuring backups. All email data sat on servers the company owned.

What is Cloud Infrastructure?

Cloud infrastructure means that instead of owning and managing physical servers, an organization subscribes to services from a cloud provider like Microsoft. The servers, storage, and applications are hosted in the provider's data centers, and the organization accesses them over the internet.

In a cloud environment:

- The cloud provider owns and maintains all physical hardware.
- The organization pays a subscription fee based on the number of users or services used.
- Applications are always up to date — Microsoft handles all updates automatically.
- Users access services from anywhere using any device and internet connection.
- Data is stored in the cloud provider's data centers with built-in backup and redundancy.

Detailed Comparison — On-Premises vs Cloud

Aspect	On-Premises	Cloud (e.g., Microsoft 365)
Hardware Ownership	Organization buys and owns all servers, storage, and networking hardware.	Cloud provider owns all hardware. Organization pays only for services used.
Upfront Cost	Very high — significant capital investment required before going live.	Low or zero — subscription-based model. Pay monthly or annually per user.
Ongoing Cost	Ongoing costs for power, cooling, hardware maintenance, replacements, and IT staff.	Predictable monthly subscription. No surprise hardware costs.
Setup Time	Weeks or months to procure hardware, install, configure, and deploy.	Hours or days — users can be onboarded quickly once subscriptions are purchased.
Maintenance	Full responsibility of the organization's IT team — patching, upgrades, monitoring.	Handled entirely by the cloud provider. IT team focuses on user management.
Scalability	Slow and expensive — requires purchasing more hardware to scale up. Downsizing means wasted investment.	Instant — add or remove users/licenses in minutes. Scale up or down as needed.
Availability	Dependent on local power, internet, and hardware. Downtime during failures.	Microsoft guarantees 99.9% uptime. Multiple data centers provide redundancy.
Disaster Recovery	Must be planned and built separately — expensive and complex.	Built in — data is automatically replicated across geographically separate data centers.

Aspect	On-Premises	Cloud (e.g., Microsoft 365)
Remote Access	Requires VPN setup and additional infrastructure for remote users.	Native — users access from anywhere with internet. No VPN needed for M365 services.
Security	Fully managed by the organization. Only as good as the internal team's capability.	Shared model — Microsoft secures infrastructure; organization manages identities and policies.
Software Updates	Manual — IT team must plan, test, and deploy updates. Often delayed due to resource constraints.	Automatic — Microsoft pushes updates seamlessly. Users always have the latest features.
Data Control	Complete physical control over data location and access.	Data in Microsoft's data centers. Data residency options available (e.g., India region).

Hybrid Approach — The Middle Ground

Many organizations do not switch entirely from on-premises to cloud overnight. Instead, they operate in a Hybrid environment — keeping some systems on-premises while moving others to the cloud. This is extremely common and is actually the most prevalent setup in large enterprises today.

Common hybrid scenarios:

- Active Directory remains on-premises, but Microsoft 365 handles email and collaboration in the cloud.
- Core financial systems remain on-premises due to compliance, while productivity tools (Teams, Outlook, OneDrive) move to Microsoft 365.
- Gradual migration — moving workloads to the cloud one at a time over a period of months or years.

> **Service Desk Context:** As a Service Desk engineer at LTM, you will primarily support users in a Hybrid or fully Cloud environment. Understanding the difference between on-premises and cloud helps you correctly diagnose issues — for example, knowing whether a user's account is managed locally (on-prem Active Directory) or in the cloud (Microsoft Entra ID), or whether an email issue lies with an on-premises Exchange Server or Exchange Online.

Topic 7: SaaS and Microsoft 365 — The Connection

We have covered cloud computing, service models, and delivery models. Now let us connect all of that directly to Microsoft 365 — so you understand exactly what kind of cloud service it is and what that means for the people who use and support it.

Microsoft 365 is a SaaS Product

Microsoft 365 is one of the world's most widely used Software as a Service (SaaS) platforms. As a SaaS product, it follows all the principles we discussed:

- Microsoft owns and operates the entire infrastructure — servers, data centers, networking, operating systems, and the applications themselves.
- Organizations subscribe to Microsoft 365 and pay per user per month or year — no hardware to buy, no servers to maintain.
- Users access their applications (Outlook, Teams, Word, Excel, SharePoint, OneDrive) from a web browser, desktop app, or mobile app — from anywhere in the world.
- Microsoft handles all updates, patches, security fixes, and feature enhancements automatically.
- The organization manages users, licenses, policies, and configurations through the Microsoft 365 Admin Center.

> **Key Point:** As a Service Desk engineer, you will never patch a Microsoft 365 server or upgrade its operating system — Microsoft does all of that. Your responsibility is to manage user accounts, assign licenses, configure settings, and troubleshoot user-facing issues.

The Shared Responsibility Model in Microsoft 365

Because Microsoft 365 is a SaaS product, both Microsoft and your organization share responsibility for keeping it secure and functional. This is called the Shared Responsibility Model.

Microsoft's Responsibility	Your Organization's Responsibility
Protects and maintains physical data centers worldwide.	Manage user identities, accounts, and access permissions.
Ensures service availability — maintains 99.9% uptime guarantee.	Define and enforce security policies (MFA, Conditional Access, etc.).
Secures the infrastructure, networking, and application platform.	Manage what data is stored, shared, and retained within the platform.
Delivers automatic updates and new features to all customers.	Assign the correct licenses to users based on their role and needs.
Provides built-in security and compliance tools for customers to use.	Configure and use the security tools properly to protect organizational data.
Maintains data redundancy and replication across multiple data centers.	Ensure users follow acceptable use policies and report security incidents.

Analogy: Think of Microsoft 365 like renting a fully furnished, managed apartment. The building owner (Microsoft) maintains the structure, lifts, electricity, and plumbing. You (the organization) decide who gets a key to which room, set the house rules, and are responsible for how the space is used.

Why Microsoft 365 Matters for Service Desk

As a Service Desk engineer, Microsoft 365 will be at the center of your daily work. Almost every user issue you encounter will touch one or more of these services:

- User cannot log in — involves Microsoft Entra ID and authentication.
- User not receiving emails — involves Exchange Online and Outlook.
- Cannot access a shared document — involves SharePoint Online or OneDrive permissions.
- Teams meeting not working — involves Microsoft Teams and network connectivity.
- New joiner needs to be set up — involves creating accounts, assigning licenses, enabling mailboxes.
- User's phone is lost and company data needs to be wiped — involves Microsoft Intune.

Section Summary: You now have a complete foundation — from understanding what the Internet is, to what a data center does, to how cloud computing works, to where Microsoft 365 fits in the cloud ecosystem. With this context, you are fully ready to go deeper into the core Microsoft 365 concepts in Section 2.

Section 2 — Core Microsoft 365 Concepts

Topic 1: Introduction to Microsoft 365

Microsoft 365 is a cloud-based productivity platform developed by Microsoft. It brings together the familiar Office applications — like Word, Excel, and PowerPoint — along with powerful collaboration tools, cloud storage, email services, and enterprise-grade security, all delivered through a single subscription.

Instead of purchasing software once and installing it on individual computers, organizations subscribe to Microsoft 365 and their users get access to everything they need — from any device, from anywhere in the world, as long as they have an internet connection.

For a Service Desk engineer, Microsoft 365 is not just a collection of apps — it is the digital workspace that every user in the organization relies on daily. Almost every support ticket you will handle will involve one or more Microsoft 365 services.

Simple Definition: Think of Microsoft 365 as your organization's digital office in the cloud. Everything a user needs to work — email, documents, meetings, storage, and communication — is available under one subscription, managed centrally, and accessible from anywhere.

What Microsoft 365 Provides

Microsoft 365 is not a single product. It is a suite of integrated services that work together seamlessly:

Service / Application	What It Does and Why It Matters
Word, Excel, PowerPoint, OneNote	The core productivity apps for creating documents, spreadsheets, presentations, and notes. Available as desktop, web, and mobile versions.
Outlook	Email client and calendar tool used by virtually every employee. Powered by Exchange Online behind the scenes.
Exchange Online	The cloud-based email hosting service. Manages mailboxes, shared mailboxes, distribution lists, and email flow for the organization.
Microsoft Teams	The primary communication and collaboration platform. Combines chat, video meetings, voice calls, file sharing, and team workspaces in one place.
SharePoint Online	Cloud-based document management and intranet platform. Stores team files, hosts internal websites, and enables content collaboration.
OneDrive for Business	Personal cloud storage for each user. Files sync across devices and can be accessed from anywhere.
Microsoft Entra ID	The cloud identity service that handles user authentication, Single Sign-On (SSO), and Multi-Factor Authentication (MFA) for all Microsoft 365 services.
Microsoft Intune	Device and application management service. Ensures only trusted, compliant devices can access company data.
Microsoft Defender for Office 365	Security service that protects against phishing emails, malicious links, and dangerous attachments.

Key Benefits of Microsoft 365

Benefit	What It Means in Practice
Work from Anywhere	Users can access their emails, files, and collaboration tools from any device with an internet connection — office, home, or on the go.
Always Up to Date	Microsoft handles all updates automatically. Users always have the latest features and security patches without IT needing to manually update software.
Subscription-Based	Organizations pay monthly or annually per user. No large upfront software costs. Licenses can be added or removed as the team grows or changes.
Built-in Security	Microsoft 365 includes identity protection, threat detection, data loss prevention, and compliance tools — all built in at no extra cost for standard plans.
Real-Time Collaboration	Multiple users can work on the same Word document, Excel sheet, or PowerPoint simultaneously — seeing each other's changes in real time.
Integrated Experience	All services are connected. A Teams meeting can access a SharePoint file. An Outlook email can schedule a Teams call. Everything works together.

The Cloud-Based Nature of Microsoft 365

Microsoft 365 is built entirely on the cloud computing model — specifically Software as a Service (SaaS), which we covered in Section 1. This means:

- Microsoft owns and operates all the servers and data centers that power these services.
- Users access everything through a browser, desktop app, or mobile app — no local servers needed.
- Microsoft guarantees 99.9% service uptime, backed by a Service Level Agreement (SLA).
- Data is automatically replicated across multiple data centers for redundancy and disaster recovery.
- Microsoft handles all infrastructure maintenance, security patching, and feature updates.

Topic 2: Licensing and Subscriptions

Understanding Microsoft 365 licensing is essential for a Service Desk engineer. When a user reports they cannot access a certain feature or application, the first thing to check is often whether they have the right license assigned. Licensing directly controls what each user can access.

How the Licensing Model Works

Microsoft 365 uses a per-user licensing model. Each user in the organization is assigned a license, and that license determines exactly which services and applications they can use. No license means no access.

Core principles of the licensing model:

- Every user needs their own license — licenses are not shared between people.
- The license assigned to a user determines what apps and services they can use — for example, whether they get desktop Office apps, Teams, Exchange, SharePoint, Intune, and so on.
- Administrators assign, change, or remove licenses through the Microsoft 365 Admin Center.
- When a license is assigned, access is granted almost immediately. When removed, access is revoked.
- Organizations only pay for the licenses they have purchased — they are billed monthly or annually.

> **Practical Example:** A new employee joins the IT Support team. The admin assigns them a Microsoft 365 Business Premium license. Within minutes, the employee has access to Outlook, Teams, Word, Excel, PowerPoint, OneDrive, SharePoint, and Intune for device management — all from a single license assignment.

Microsoft 365 Subscription Plans

Microsoft offers different subscription plans depending on the size of the organization and its requirements. These fall into three broad categories:

A. Business Plans — For Organizations up to 300 Users

Plan	What's Included
Microsoft 365 Business Basic	Web and mobile Office apps only (no desktop apps). Email via Exchange Online. Teams, OneDrive, and SharePoint. Suitable for light users who primarily work on the web.
Microsoft 365 Business Standard	Everything in Business Basic, plus full desktop Office apps (Word, Excel, PowerPoint, etc.). Additional productivity features. Suitable for most office workers.
Microsoft 365 Business Premium	Everything in Business Standard, plus advanced security tools, Intune for device management, and Azure AD Premium P1 for enhanced identity features. Best for organizations needing stronger security.
Microsoft 365 Apps for Business	Desktop Office applications only — no email, Teams, or cloud services included. Suitable for users who only need the Office apps installed.

B. Enterprise Plans — For Large Organizations (No User Limit)

Plan	What's Included
Microsoft 365 E1	Web and mobile Office apps. Teams, SharePoint, and OneDrive. No desktop Office apps. Entry-level enterprise plan.
Microsoft 365 E3	Everything in E1, plus full desktop Office apps and additional security and compliance tools. The most commonly deployed enterprise plan.
Microsoft 365 E5	Everything in E3, plus advanced security features, Microsoft Defender, advanced threat protection, enhanced compliance and analytics. The most comprehensive plan available.

C. Standalone Plans

Some organizations do not need the full suite — they purchase only the specific services they require. Examples include Exchange Online only (for email), SharePoint Online only (for document management), Teams Phone (for calling), or Defender for Office 365 (for security). These can be purchased individually or added on top of an existing plan.

Feature Comparison Across Common Plans

Feature	Biz Basic	Biz Standard	Biz Premium	E3	E5
Desktop Office Apps (Word, Excel, etc.)	No	Yes	Yes	Yes	Yes

Feature	Biz Basic	Biz Standard	Biz Premium	E3	E5
Email — Exchange Online	Yes	Yes	Yes	Yes	Yes
Teams, OneDrive, SharePoint	Yes	Yes	Yes	Yes	Yes
Intune — Device Management	No	No	Yes	Yes	Yes
Azure AD Premium P1	No	No	Yes	Yes	Yes
Advanced Security & Compliance	No	No	Medium	Medium	High
Microsoft Defender for Office 365	No	No	Add-On	Add-On	Included

Understanding License Components — Service Plans

Every Microsoft 365 license is made up of smaller components called Service Plans. For example, a Microsoft 365 E3 license includes Exchange Online, SharePoint Online, Teams, OneDrive, Intune, and more — all bundled together. Administrators can enable or disable individual service plans within a license for specific users.

This means a user can have an E3 license but have certain services switched off — for example, disabling Teams for a specific role that does not need it, while keeping all other services active.

Add-On Licenses

Add-on licenses allow organizations to extend specific features for users without upgrading their entire subscription plan. Common add-ons include:

- Teams Phone — Adds calling capabilities to Teams (replacing traditional desk phones).
- Audio Conferencing — Allows users to join Teams meetings via a phone dial-in number.
- Microsoft Defender for Office 365 — Advanced email security, can be added to Basic or Standard plans.
- Power BI Pro — Advanced data analytics and reporting.
- Viva modules — Employee experience tools for learning, insights, and engagement.

> **Service Desk Tip:** When a user says they cannot find a feature — for example, they cannot see the desktop version of Word, or they cannot use Intune on their device — the first step is to check their assigned license in the Admin Center. The license tells you exactly what they should and should not have access to.

Topic 3: Identity and Access Management

Identity and Access Management (IAM) is the foundation of security in Microsoft 365. Every time a user signs in, accesses a file, sends an email, or joins a Teams meeting — their identity is verified and their level of access is determined. Without a properly managed identity, nothing in Microsoft 365 works.

In Microsoft 365, identity management is handled by Microsoft Entra ID — previously known as Azure Active Directory (Azure AD). Entra ID is the cloud-based directory service that sits behind every Microsoft 365 sign-in.

> **Simple Way to Think About It:** Entra ID is the digital security gate of Microsoft 365. Every user, administrator, and application must pass through this gate — Entra ID checks who you are, whether your credentials are valid, and what you are allowed to access before letting you in.

Types of User Accounts in Microsoft 365

Microsoft 365 supports three types of user identities, each with a different level of access:

Account Type	Who Uses It	Key Characteristics
Standard User Account	Regular employees	Assigned a license. Can access apps and services based on their subscription. Cannot manage other users or admin settings. This is the most common account type.
Administrator Account	IT administrators	Has elevated privileges to manage the Microsoft 365 environment. Different admin roles exist — from Global Admin (full control) to service-specific roles like Exchange Admin or Teams Admin.

Account Type	Who Uses It	Key Characteristics
Guest User Account	External collaborators	Invited from outside the organization to collaborate on Teams, SharePoint, or shared documents. Access is controlled and limited to what the organization permits.

How User Accounts are Created and Managed

User accounts in Microsoft 365 can be created and managed in several ways depending on the organization's size and environment:

- Manual Creation — Admins create individual accounts through the Microsoft 365 Admin Center. Suitable for small teams or occasional new hires.
- Bulk Import — Multiple accounts can be created at once using a CSV file. Useful when onboarding a large number of users simultaneously — for example, when a new batch of employees joins.
- Directory Synchronization — In organizations that have on-premises Active Directory, user accounts are synchronized to Microsoft 365 using a tool called Azure AD Connect (now called Entra Connect). This ensures users have the same identity on-premises and in the cloud.

Once created, administrators can manage user account details such as:

- Contact information — name, phone number, job title, department.
- Locale settings — language, time zone, and regional preferences.
- Role assignments — what administrative permissions the user has.
- License assignments — which Microsoft 365 services the user can access.

Authentication Methods

Authentication is the process of verifying a user's identity before granting access to Microsoft 365 services. The platform supports several authentication methods:

Authentication Method	How It Works and When It Is Used
Username and Password	The basic form of authentication. User enters their email address and password. While simple, this alone is considered insufficient for securing corporate accounts today.
Single Sign-On (SSO)	Allows a user to authenticate once and gain access to multiple applications without being prompted to log in again. Example: A user signs in to their Windows PC (which is joined to the domain) and automatically gets access to Outlook, Teams, SharePoint, and other M365 services without re-entering credentials.
Multi-Factor Authentication (MFA)	Adds a second layer of verification on top of the password. After entering their password, the user must also verify their identity using a phone call, SMS code, or an authenticator app. Even if a password is stolen, MFA prevents unauthorized access.
Passwordless Authentication	Removes the password entirely. Users authenticate using the Microsoft Authenticator app, a FIDO2 security key, or Windows Hello biometrics. Increasingly common in modern security-conscious environments.

> **Why MFA Matters:** Over 99% of account compromise attacks can be blocked simply by enabling MFA. It is the single most effective security measure an organization can implement. As a Service Desk engineer, you will frequently assist users who are locked out due to MFA issues — knowing how MFA works is essential.

Self-Service Password Reset (SSPR)

Self-Service Password Reset (SSPR) is a feature that allows users to reset their own passwords without calling the Service Desk. This significantly reduces the number of password reset tickets — one of the most common Service Desk requests.

How SSPR works:

- The user goes to the Microsoft 365 sign-in page and clicks 'Forgot my password'.
- They are prompted to verify their identity using pre-registered methods — such as an alternate email, phone number, or authenticator app.
- Once verified, they set a new password immediately and regain access.

Key points about SSPR:

- Administrators configure which verification methods are allowed and how many are required.
- Users must register their verification methods before they can use SSPR — this is typically done during initial onboarding.
- SSPR reduces IT workload and gives users a faster, 24/7 way to recover their accounts.

> **Service Desk Context:** Even with SSPR in place, you will still receive password-related tickets — usually from users who have not registered their SSPR methods, users locked out due to too many failed attempts, or users who need their MFA authenticator reset because they changed their phone. These are very common scenarios.

Conditional Access

Conditional Access is one of the most powerful security features in Microsoft 365. It allows administrators to define rules that control when and how users can access Microsoft 365 services — based on specific conditions at the time of sign-in.

Instead of a simple yes/no access decision, Conditional Access evaluates a set of signals and then grants access, blocks it, or requires additional verification.

Conditions that Conditional Access can evaluate:

- User identity — who is signing in (regular user, admin, guest)?
- Device compliance — is the device enrolled in Intune and meeting security standards?
- Location — is the sign-in coming from a trusted office network or an unknown country?
- Application — which Microsoft 365 service is being accessed?
- Sign-in risk — has Microsoft's AI detected unusual login behavior?

Based on these conditions, Conditional Access can:

- Grant access — everything looks normal, allow the user in.
- Require MFA — the sign-in looks slightly risky, ask for a second verification.
- Block access entirely — the sign-in is from a blocked country or a non-compliant device.

> **Example:** A Conditional Access policy is configured that says: 'If a user tries to access Microsoft 365 from outside India, require MFA.' An employee on a business trip to Singapore signs in — they are prompted for MFA. A bad actor in another country trying to access the same account with a stolen password is blocked unless they can also complete MFA.

Managing Roles in Microsoft 365

Microsoft 365 uses a Role-Based Access Control (RBAC) model to manage administrative permissions. Instead of giving all IT staff full administrator access, specific roles are assigned based on what each person needs to do.

Admin Role	What They Can Do
Global Administrator	Full control over all Microsoft 365 services and settings. Can manage everything. Should be limited to a very small number of trusted individuals.
User Administrator	Can create, edit, and delete user accounts. Can manage passwords, licenses, and basic user settings. Common Service Desk role.
Exchange Administrator	Manages email settings — mailboxes, shared mailboxes, distribution lists, mail flow rules, and Exchange Online configurations.
Teams Administrator	Manages Teams policies, meeting settings, calling features, and Teams-specific user configurations.
SharePoint Administrator	Manages SharePoint sites, external sharing settings, storage, and access permissions.
Helpdesk Administrator	Can reset passwords and manage basic user settings, but cannot make broader configuration changes. Designed specifically for front-line support.
Security Administrator	Manages security policies, threat protection settings, and security-related configurations across Microsoft 365.

> **Principle of Least Privilege:** Always assign the minimum role needed for a person to do their job. A Service Desk engineer should have the Helpdesk Administrator or User Administrator role — not Global Administrator. If Global Admin credentials are compromised, the entire Microsoft 365 environment is at risk.

Topic 4: Microsoft 365 Administration

Microsoft 365 administration involves managing the entire Microsoft 365 environment — users, licenses, groups, security policies, service configurations, and more. Administrators use a set of dedicated online portals called Admin Centers to perform these tasks. As a Service Desk engineer, you will spend a significant portion of your day inside these portals.

The Microsoft 365 Admin Center

The Microsoft 365 Admin Center (admin.microsoft.com) is the central management portal for the entire Microsoft 365 environment. It is the first place an administrator goes for most tasks.

Key tasks performed in the Microsoft 365 Admin Center:

- Creating, modifying, and deleting user accounts.
- Assigning and removing licenses.
- Managing groups — Microsoft 365 groups, distribution lists, and security groups.
- Configuring organizational settings such as the company name, domain, and branding.
- Monitoring service health — checking if any Microsoft 365 services are experiencing issues.
- Viewing usage reports and activity dashboards.
- Managing billing and subscriptions.

Specialized Admin Centers

In addition to the main Admin Center, Microsoft 365 provides separate, specialized portals for managing specific services. Each service has its own dedicated admin center with deeper controls:

Admin Center	What It Manages
Exchange Admin Center	Manages all email-related settings — user mailboxes, shared mailboxes, distribution lists, mail flow rules, anti-spam settings, and email routing.
Teams Admin Center	Controls Microsoft Teams policies, meeting settings, calling features, Teams app permissions, and user-specific Teams configurations.
SharePoint Admin Center	Manages SharePoint sites, hub sites, external sharing settings, storage quotas, and site-level permissions.
Microsoft Intune Admin Center	Manages device enrollment, compliance policies, app protection policies, and remote actions for enrolled devices.
Microsoft Purview (Compliance Center)	Handles Data Loss Prevention (DLP) policies, retention policies, eDiscovery, audit logs, communication compliance, and data governance.
Microsoft Entra Admin Center	Manages identities, user accounts, groups, Conditional Access policies, MFA settings, and directory synchronization.
Microsoft 365 Defender	Centralized security portal for monitoring threats, reviewing security alerts, and managing Microsoft Defender policies.

> **Practical Note:** You do not need to memorize every function of every admin center right now. What is important is knowing which portal to go to for which type of task. If a user has an email issue — Exchange Admin Center. If a user is locked out — Entra Admin Center. If a device needs to be wiped — Intune Admin Center.

Monitoring Service Health

One of the most important tasks in Microsoft 365 administration is monitoring whether Microsoft's services are running correctly. Microsoft provides a Service Health Dashboard inside the Admin Center that shows the real-time status of all Microsoft 365 services.

The Service Health Dashboard shows:

- Current status of each service — whether it is healthy, degraded, or experiencing an outage.
- Active incidents — descriptions of ongoing issues, their impact, and Microsoft's progress in resolving them.
- Planned maintenance — scheduled maintenance windows that may affect service availability.
- Historical data — past incidents and how they were resolved.

Why this matters for Service Desk: When multiple users report the same issue — for example, 'Teams is not working' or 'I cannot access my email' — the first thing to check is the Service Health Dashboard. If Microsoft has a known outage, the resolution is on Microsoft's side and users simply need to wait. This saves time and avoids unnecessary troubleshooting.

Usage Reports and Activity Monitoring

Microsoft 365 provides detailed usage reports that give administrators visibility into how the organization is using the platform. These reports are available in the Microsoft 365 Admin Center under the Reports section.

Common reports available:

- Email activity — how many emails are being sent and received across the organization.
- Teams usage — active users, meetings held, messages sent, and channel activity.
- OneDrive and SharePoint activity — files stored, shared, and synced.
- License usage — which licenses are assigned versus which are actually being actively used.
- Security reports — sign-in activity, risky sign-ins, and MFA usage statistics.

> **Cost Optimization:** License usage reports are particularly valuable. If a large number of assigned licenses are showing no activity, the organization may be paying for licenses that are not being used. These can be reclaimed and either removed or reassigned to new users, saving cost.

Topic 5: Mailboxes and Groups in Exchange Online

Exchange Online is the cloud-based email and calendaring service that powers Outlook in Microsoft 365. It is one of the most heavily used services in the platform and one of the most common areas of Service Desk activity. Understanding the different types of mailboxes and groups — and when each is used — is essential for your role.

What is Exchange Online?

Exchange Online is Microsoft's cloud-hosted email server. In traditional on-premises environments, organizations ran Microsoft Exchange Server on their own physical hardware. Exchange Online moves this entirely to Microsoft's cloud — giving organizations business-class email without the burden of maintaining their own mail server.

Exchange Online provides:

- Business email hosting for all users in the organization.
- Calendar and scheduling features, including shared calendars and resource booking.
- Shared mailboxes for team-based email accounts (e.g., support@company.com).
- Distribution lists and Microsoft 365 groups for sending emails to groups of people.
- Anti-spam and anti-malware filtering through Exchange Online Protection (EOP).
- Email flow management — controlling how emails are routed, filtered, and delivered.

Types of Mailboxes

Exchange Online supports several types of mailboxes, each serving a different purpose. Knowing which type to use and when is a key Service Desk skill:

User Mailbox

A User Mailbox is the standard email account assigned to an individual employee. Each user mailbox is tied to a specific person's account and requires a Microsoft 365 license that includes Exchange Online. This is the most common mailbox type — every employee who needs an email address gets a user mailbox.

- Requires a license — no license means no mailbox.
- The user has full, exclusive access to their own mailbox.
- Has its own email address (e.g., priya.sharma@ltm.com).
- Includes email, calendar, contacts, and tasks.

Shared Mailbox

A Shared Mailbox is an email account that multiple users can access and send emails from, without requiring a separate license. Shared mailboxes are used for team or department email addresses where multiple people need to monitor and respond to incoming emails.

- Does not require its own license (up to 50GB of storage).

- Multiple users are granted access by an administrator.
- Users can send emails as the shared mailbox address or on behalf of it.
- Ideal for: support@company.com, helpdesk@company.com, info@company.com.
- Emails sent to the shared mailbox are visible to all members who have access.

> **Common Service Desk Task:** A new support team is created and needs a shared email address that all team members can monitor. The admin creates a shared mailbox (e.g., it-support@ltm.com), grants all team members access, and they can all read and respond to incoming emails from that address.

Resource Mailbox — Room Mailbox

A Room Mailbox represents a physical meeting room that can be booked through Outlook or Teams. When a user schedules a meeting and wants to book a conference room, they add the room mailbox as a location. The room mailbox can be configured to auto-accept or require manual approval for booking requests.

- Represents a physical room (e.g., 'Mumbai Conference Room A').
- Users see real-time availability when scheduling meetings.
- Can be set to auto-accept bookings if the room is free.
- Does not require a license — it is a resource, not a person.

Resource Mailbox — Equipment Mailbox

An Equipment Mailbox works similarly to a room mailbox but represents a piece of equipment rather than a room. Organizations use equipment mailboxes for shared items like projectors, cameras, company vehicles, or audio systems that need to be booked in advance.

- Examples: 'Projector - Floor 3', 'Company Car - MH01', 'Video Camera - Studio'.
- Bookable through Outlook calendar just like a room mailbox.
- Does not require a license.

Types of Groups in Microsoft 365

Groups in Microsoft 365 are used to bring users together — either for communication, collaboration, or access control. There are several types of groups, and each serves a different purpose. Choosing the right group type is important — using the wrong one leads to confusion and extra administrative work.

Microsoft 365 Group

A Microsoft 365 Group is a modern collaboration group that automatically creates a set of shared resources for a team. When you create a Microsoft 365 Group, you get a shared mailbox and email address, a shared calendar, a SharePoint document library, a Planner board, and optionally a Teams workspace — all connected and accessible to every group member.

- Ideal for ongoing teams and projects that need all-round collaboration.
- All members can access the shared mailbox, files, and calendar.
- When you create a Team in Microsoft Teams, a Microsoft 365 Group is automatically created behind the scenes.
- Membership can be managed by the group owner or by an administrator.

> **Example:** A new 'Product Launch' project team is created. The admin creates a Microsoft 365 Group called 'Product Launch Team'. All members instantly get a shared mailbox (productlaunch@ltm.com), a shared SharePoint site for documents, a shared calendar for milestones, and a Teams channel for daily communication.

Distribution Group

A Distribution Group (also called a Distribution List or DL) is a simple email list. It has an email address, and when someone sends an email to that address, it is delivered to all members of the list. It does not create any shared resources — no shared files, no calendar, no Teams workspace.

- Used purely for sending emails to multiple people at once.
- The recipients cannot reply to each other via the group — it is one-way communication.
- Common uses: company announcements, newsletter lists, team broadcast emails.
- No shared storage, no collaboration features — email distribution only.

> **Example:** HR wants to send a company-wide announcement. They send it to all-staff@ltm.com — a distribution list that includes every employee. The email lands in each person's individual inbox. No one's replies go to the group — it is a broadcast, not a conversation.

Mail-Enabled Security Group

A Mail-Enabled Security Group combines the functions of a security group (used for assigning permissions to resources like SharePoint sites, shared drives, or applications) with the ability to receive email. It has both an email address for communication and security access for permission assignments.

- Can be emailed like a distribution list.
- Can also be assigned permissions to resources — like granting access to a SharePoint site.
- Useful when a team needs both an email address and access to specific shared resources.

> **Example:** The HR team has a Mail-Enabled Security Group called 'HR Team' with email hr@ltm.com. This group is used to grant all HR members access to the HR SharePoint library (permissions), and employees can email the entire HR team at hr@ltm.com (email distribution) — both from the same group.

Dynamic Group

A Dynamic Group automatically manages its own membership based on rules defined by an administrator. Instead of manually adding or removing users, the system looks at user attributes in Entra ID — such as department, job title, location, or employment type — and automatically adds or removes users as their attributes change.

- Membership is automatic — no manual add or remove needed.
- Rules are defined using user attributes (e.g., Department = 'Finance').
- Requires Azure AD Premium (P1 or P2) licensing.
- Extremely useful in large organizations where staff movements are frequent.

> **Example:** An admin creates a Dynamic Group with the rule: 'All users where Department = Sales'. Every time a new employee joins the Sales team, their account is automatically added to this group — and when they transfer to another department, they are automatically removed. No manual membership management required.

Quick Reference — Group Types

Group Type	Has Email?	Has Shared Resources?	Best Used For
Microsoft 365 Group	Yes	Yes — SharePoint, Calendar, Planner, Teams	Ongoing team collaboration and projects
Distribution Group	Yes	No — email only	Broadcast emails to a list of people
Mail-Enabled Security Group	Yes	No — permissions + email only	Teams needing both email and resource access
Dynamic Group	Optional	Depends on type	Auto-managing membership based on attributes
Security Group	No	No — permissions only	Controlling access to apps, sites, and files

Topic 6: Microsoft Teams

Microsoft Teams is the central hub for communication and collaboration in Microsoft 365. It replaces the need for separate tools for chat, video conferencing, file sharing, and project collaboration — bringing everything together in one integrated platform.

For most organizations, Teams has become the primary way employees communicate daily — replacing traditional phone calls, emails for quick conversations, and in many cases, physical meeting rooms. As a Service Desk engineer, you will use Teams yourself and support users who rely on it every day.

What is Microsoft Teams?

Microsoft Teams is a collaboration platform that combines persistent chat, video and audio meetings, file sharing, and app integrations into a single workspace. It is deeply integrated with the rest of Microsoft 365 — files shared in Teams are stored in SharePoint, meetings are scheduled through the Outlook calendar, and identities are managed through Entra ID.

Core capabilities of Microsoft Teams:

- Chat — One-on-one and group text conversations with persistent history.
- Channels — Organized discussion spaces within a team, grouped by topic or project.

- Meetings — Video and audio meetings with screen sharing, recording, and transcription.
- Calling — Voice calls between individuals, replacing traditional desk phones in many organizations.
- File Sharing — Upload, share, and co-author documents directly within Teams.
- Apps and Integrations — Add third-party apps, tabs, and bots to extend functionality.

Teams and Channels — The Structure

Understanding the structure of Teams is important for both using it and supporting users who have questions about it.

What is a Team?

A Team is a workspace in Microsoft Teams created for a specific group of people — usually a department, project, or functional area. It acts as the top-level container that holds all the related channels, files, and conversations for that group.

- Examples: 'IT Support Team', 'HR Department', 'Project Apollo'.
- Every Team is backed by a Microsoft 365 Group — which means it also has a shared mailbox, SharePoint site, and calendar.
- Teams can be Public (anyone in the organization can join) or Private (membership by invitation only).
- A Team can have multiple channels organized by topic.

What is a Channel?

A Channel is a focused discussion space within a Team. Channels help organize conversations by topic, project, or workstream so that discussions do not get mixed up. Every Team starts with a General channel by default.

Channel Type	Description	Example
Standard Channel	Open to all members of the Team. All conversations and files are visible to every team member.	'Announcements' or 'General' channel in the IT Support Team — visible to all.
Private Channel	Restricted to specific members within the Team. Even other Team members cannot see the conversations or files in a Private Channel.	'Escalations' channel visible only to senior engineers, hidden from others.
Shared Channel	Can include people from outside the Team — even from outside the organization — without making them full Team members.	A 'Client Collaboration' channel shared with an external vendor or client.

Meetings in Microsoft Teams

Teams meetings are one of the most heavily used features in Microsoft 365, especially with the rise of hybrid and remote work. Teams supports several types of meetings:

Meeting Type	Description and Use Case
Scheduled Meeting	A meeting created in advance through Teams or Outlook calendar. Attendees receive a calendar invite with a link to join. Used for planned discussions, project reviews, and recurring stand-ups.
Meet Now	An instant, unscheduled meeting started immediately from a chat or channel. No calendar invite needed. Used for quick, ad-hoc discussions.
Webinar	A structured presentation to a larger audience. Presenters can be managed separately from attendees. Attendees may be restricted from speaking or sharing.
Live Event	A broadcast-style event for very large audiences — hundreds or thousands of viewers. Used for all-hands meetings, town halls, and company-wide announcements.
Channel Meeting	A meeting held within a specific channel. All channel members can see it, join it, and access the recording afterward. Transparent to the entire team.

File Sharing in Teams

Files shared in Microsoft Teams are not just temporary attachments — they are stored permanently in SharePoint behind the scenes. This means:

- Every Team has its own SharePoint document library where team files are stored.
- Files shared in a channel are stored in the corresponding SharePoint folder for that channel.

- Files shared in a private chat are stored in the sender's OneDrive.
- Multiple users can co-author (edit simultaneously) the same document within Teams.
- Files remain accessible even if the person who originally uploaded them leaves the organization.

> **Important for Service Desk:** When a user says 'I lost a file I shared in Teams' — it is stored in SharePoint. When they say 'I cannot find a file someone sent me in a chat' — it is in their OneDrive or the sender's OneDrive. Knowing where Teams stores files helps you direct users to the right place.

Teams Policies and Settings

Microsoft Teams behaviour is governed by policies configured in the Teams Admin Center. These policies control what users can and cannot do within Teams:

- Meeting Policies — Control recording permissions, who can present, whether external users can join, and lobby bypass settings.
- Messaging Policies — Control whether users can delete or edit messages, use GIFs, or send urgent notifications.
- App Permission Policies — Control which Teams apps (including third-party apps) users are allowed to install and use.
- Calling Policies — Control voice calling features, voicemail, and call forwarding options.

> **Service Desk Context:** If a user reports they cannot record a meeting, cannot add an external guest to a Team, or cannot use a specific app in Teams — the cause is almost always a policy restriction. The Teams Admin Center is where these policies are configured and reviewed.

Topic 7: SharePoint Online and OneDrive

SharePoint Online and OneDrive for Business are the two primary file storage and management services in Microsoft 365. While they both store files in the cloud, they serve fundamentally different purposes — and understanding this difference is one of the most common areas of confusion for both users and new Service Desk engineers.

> **Simple Distinction:** OneDrive is your personal cloud storage — like your own private desk drawer. SharePoint is team or organizational storage — like a shared filing cabinet that the whole team has access to.

SharePoint Online

SharePoint Online is a cloud-based platform for team collaboration, document management, and building internal websites. It is the backbone of team file storage in Microsoft 365 — every Microsoft Teams team and every Microsoft 365 Group has a SharePoint site behind it, even if users do not always realize it.

What SharePoint Online is used for:

- Storing and organizing team documents and project files in a shared, structured way.
- Building internal websites — company intranets, department portals, and knowledge hubs.
- Managing content with features like document versioning, check-in/check-out, and metadata.
- Controlling who can access what — owners, members, and visitors with different permission levels.
- Collaborating on documents in real time with co-authoring.

Types of SharePoint Sites

Site Type	Purpose	Typical Use
Team Site	A collaborative workspace for a specific team or project. Includes a document library, shared calendar, and notebook.	An IT Support team's SharePoint site for storing runbooks, SOPs, and project documents.
Communication Site	A broadcasting site for sharing news, announcements, and information with a wide audience. Read-heavy — most visitors only read.	A company intranet homepage with HR announcements, company news, and leadership updates.
Hub Site	A parent site that connects multiple related team or communication sites together for unified navigation and search.	An 'HR Hub' that connects the HR Team site, the Policies site, and the Training site under one umbrella.

SharePoint Permissions

Access to SharePoint sites is managed through permission levels. Understanding these helps you support users who report they cannot access a file or site:

Permission Level	Who Has It	What They Can Do
Owner	Site owners (usually team leads or admins)	Full control — can manage settings, permissions, add or remove members, and delete the site.
Member	Regular team members	Can read, create, edit, and delete files and content within the site. Most team members are at this level.
Visitor	External or read-only users	Can view and read content only. Cannot create, edit, or delete anything.

> **Service Desk Context:** The most common SharePoint-related support request is 'I cannot access this site or file.' The answer almost always comes down to permissions. Check what permission level the user has (or if they have been added to the site at all) before escalating.

Document Versioning

SharePoint automatically keeps a history of changes to documents when versioning is enabled. Every time someone edits and saves a file, a new version is created and the previous version is preserved. This means:

- Users can view the full history of changes to a document.
- Previous versions can be restored if a mistake is made or content is accidentally deleted.
- Admins can configure how many versions to keep — for example, the last 500 versions.

> **Example:** A team member accidentally deletes an important section from a Word document stored in SharePoint and saves it. The document owner can go to Version History, see the version before the change was made, and restore it — recovering the lost content completely.

OneDrive for Business

OneDrive for Business is each user's personal cloud storage space within Microsoft 365. Every licensed user gets their own dedicated OneDrive storage — separate from team storage — for files that are personal to their work but need to be accessible from any device.

Key characteristics of OneDrive:

- Each user gets their own private OneDrive space — other people cannot see it unless the user shares specific files or folders.
- Files stored in OneDrive sync automatically to the user's devices — so they are available offline too.
- Users can share individual files or folders with specific colleagues when needed.
- OneDrive is the storage behind files shared in private Teams chats.
- Default storage is 1TB per user.

Common uses of OneDrive:

- Storing work-in-progress files that are not ready to be shared with the team yet.
- Personal reference documents, templates, and working drafts.
- Files that belong to an individual rather than a team.
- Syncing files for offline access when working without internet connectivity.

SharePoint vs OneDrive — When to Use Which

Use SharePoint When...	Use OneDrive When...
The file belongs to a team, not an individual.	The file is personal and only you need access to it.
Multiple people need to access, edit, or collaborate on the document.	You are working on a draft and are not ready to share it yet.

Use SharePoint When...	Use OneDrive When...
The content needs to be organized in a structured team library.	You need to share a specific file with one or two colleagues temporarily.
The file should remain accessible even if one person leaves.	You want to sync files to your laptop for offline access.
You are building a team intranet or knowledge base.	You need personal cloud storage accessible from any device.

Topic 8: Outlook — Email and Calendar Management

Microsoft Outlook is the primary email and calendar application used by employees in Microsoft 365. It connects to Exchange Online for email hosting and provides a rich set of features for managing communication, scheduling, and contacts. For a Service Desk engineer, Outlook-related issues are among the most frequent tickets you will handle.

Outlook — Overview

Outlook is available in three forms, all connected to the same Exchange Online mailbox:

- Outlook Desktop App — Installed on the user's Windows or Mac computer. The richest feature set and preferred by most users. Part of Microsoft 365 subscriptions that include desktop apps.
- Outlook on the Web (OWA) — Accessed directly from a browser at outlook.office.com. No installation required. Useful for troubleshooting — if the desktop app has issues but OWA works, the problem is likely the local installation.
- Outlook Mobile App — Available for iOS and Android. Lets users manage email and calendar from their smartphones.

> **Troubleshooting Tip:** When a user says 'Outlook is not working', always ask them to try Outlook on the Web (OWA) first. If OWA works, the issue is with the desktop application or local configuration — not with their mailbox or Exchange Online. This simple check narrows down the root cause quickly.

Email Management in Outlook

Outlook provides a comprehensive set of tools for managing email efficiently. Key features include:

Feature	What It Does and Why It Matters
Focused Inbox	Automatically sorts emails into two tabs — Focused (important emails) and Other (newsletters, notifications). Helps users prioritize important messages.
Folders	Users can create custom folders to organize emails by project, sender, or topic. Good email hygiene prevents cluttered inboxes.
Categories	Color-coded labels that can be applied to emails for quick visual organization. Categories can also be applied to calendar events and tasks.
Flags	Mark an email with a flag as a reminder to follow up on it. Flagged items appear in the Tasks pane as well.
Rules	Automated actions triggered by specific conditions — for example, automatically moving all emails from a certain sender to a specific folder, or forwarding certain emails to a colleague.
Search	Outlook's search scans across email, calendar, contacts, and attachments. Users can filter by sender, date, subject, or keyword.
Email Signatures	Users can set up professional email signatures that are automatically appended to every outgoing email.

Calendar Management in Outlook

The Outlook Calendar is tightly integrated with Microsoft Teams and Exchange Online. It is used not just for personal scheduling, but for team coordination across the organization.

Key calendar features:

- Creating appointments and meetings — Schedule personal time blocks or invite others to meetings.

- Scheduling Assistant — When creating a meeting invite, the Scheduling Assistant shows the availability of all invited attendees, making it easy to find a time that works for everyone.
- Room Booking — Users can add a meeting room (Resource Mailbox) directly from the calendar when creating a meeting invite. Outlook shows whether the room is available.
- Recurring Meetings — Create meetings that repeat daily, weekly, monthly, or on a custom schedule.
- Calendar Sharing — Users can share their calendar with colleagues to show their availability, helping with scheduling and planning.
- Delegate Access — A user can grant another person (typically an executive assistant) full access to manage their calendar and emails on their behalf.

> **Common Scenario:** An executive assistant manages the calendar for a senior manager. The admin grants the assistant 'delegate access' to the manager's mailbox — allowing the assistant to read, create, and respond to meeting invites on the manager's behalf. This is a very common Service Desk request.

Automatic Replies (Out of Office)

Automatic Replies — commonly known as Out of Office (OOO) — allow users to set up an automated response that is sent to anyone who emails them during a specified period. This is typically used when an employee is on leave, traveling, or unavailable.

Key features of Automatic Replies:

- Users can set a specific date and time range — the auto-reply activates and deactivates automatically.
- Different messages can be configured for people inside the organization versus external senders.
- The message can include details such as the return date, an alternative contact, and urgent escalation instructions.

Email Rules and Automation

Outlook Rules are one of the most powerful email management tools available to users. A rule is an automated action that Outlook takes on incoming or outgoing emails based on conditions you define.

Examples of common email rules:

- Move all emails from a specific sender (e.g., alerts@monitoring.com) to a dedicated 'Alerts' folder automatically.
- Flag all emails that mention a specific keyword (e.g., 'Urgent') for immediate follow-up.
- Automatically forward emails from a specific project team to a shared mailbox.
- Delete emails from known spam senders before they even reach the inbox.

Outlook Profiles and Account Configuration

An Outlook Profile is the local configuration that connects the Outlook desktop app to a user's Exchange Online mailbox. Each profile contains the account credentials, server settings, and local cache data. For Service Desk engineers, understanding profiles is important because corrupted profiles are a common cause of Outlook issues.

Common Outlook issues caused by profile problems:

- Outlook stuck at 'Loading Profile' when opening.
- Outlook showing as 'Disconnected' or 'Trying to connect' in the status bar.
- Calendar and emails not syncing or showing outdated data.
- Outlook crashing or freezing repeatedly.

> **Service Desk Tip:** If a user's Outlook desktop app is consistently having issues that OWA does not have, the solution is often to create a new Outlook profile. This rebuilds the local connection to the mailbox from scratch and resolves most configuration-related issues.

Topic 9: Microsoft Office Apps

Microsoft Office Applications — commonly known simply as Office Apps — are the productivity tools that come with Microsoft 365. They are among the most widely used software applications in the world and form the core of most employees' daily work. For a Service Desk engineer, understanding what each app does and its key features is important for supporting users effectively.

How Office Apps are Delivered

Microsoft 365 Office Apps are available in three forms, depending on the subscription plan and the user's needs:

Type	Description	Best For
Desktop Apps	Fully installed on the user's computer. Richest feature set, works offline, and can handle large or complex files. Available in Microsoft 365 plans that include desktop apps (Standard, Premium, E3, E5).	Power users who work with complex documents, spreadsheets, or presentations.
Web Apps	Accessed directly in a browser at office.com. No installation needed. Slightly fewer features than desktop but suitable for most everyday tasks.	Users on shared computers, remote access, or basic editing needs.
Mobile Apps	Downloaded to iOS or Android smartphones and tablets. Optimized for touch and mobile use. Supports viewing, editing, and sharing on the go.	Users who need to review or edit documents while mobile.

Microsoft Word

Microsoft Word is the go-to application for creating text-based documents — reports, letters, proposals, contracts, and more. It is the most widely used word processing software in professional environments.

Key capabilities:

- Creating and formatting professional documents with headings, styles, tables, and images.
- Track Changes — allows reviewers to suggest edits without directly modifying the original text. The author can then accept or reject each change. Essential for collaborative document review.
- Comments — reviewers can add notes and feedback without changing the document text itself.
- Mail Merge — automatically generates personalized copies of a document for multiple recipients using a data source (e.g., a list of names and addresses).
- Templates — pre-built document layouts for common formats like invoices, resumes, and reports.
- Real-Time Co-Authoring — multiple users can edit the same document simultaneously when it is stored in SharePoint or OneDrive.

Microsoft Excel

Microsoft Excel is the leading spreadsheet application for data entry, analysis, calculation, and reporting. It is widely used across every department — from finance and HR to operations and IT.

Key capabilities:

- Organizing data in rows, columns, and tables for easy management and analysis.
- Formulas and Functions — built-in calculations like SUM, AVERAGE, VLOOKUP, IF, and COUNTIF that automate data analysis without writing code.
- PivotTables — a powerful tool that allows users to summarize and analyze large datasets by grouping, filtering, and aggregating data in seconds.
- Charts and Graphs — visualize data as bar charts, line graphs, pie charts, and more.
- Conditional Formatting — automatically highlight cells based on their values (e.g., color all values above a threshold in red).
- Data Validation — restrict what type of data can be entered in a cell, reducing errors.

Microsoft PowerPoint

Microsoft PowerPoint is the standard tool for creating presentations. It is used for business presentations, training sessions, project proposals, and any scenario where visual communication is needed.

Key capabilities:

- Slide-based presentations with text, images, charts, and multimedia content.
- Design Themes and Templates — pre-designed visual layouts that give presentations a consistent, professional look.
- Animations and Transitions — add visual effects to individual elements or slide changes.
- Presenter View — while the audience sees the presentation, the presenter sees a private view with speaker notes, a timer, and upcoming slides.
- Export to PDF or Video — presentations can be saved as PDFs or converted to video files for sharing.
- Co-Authoring — multiple users can work on the same presentation simultaneously.

Microsoft OneNote

Microsoft OneNote is a digital notebook application for capturing notes, ideas, meeting minutes, and reference material. Unlike Word, OneNote is free-form — you can place text, images, links, and drawings anywhere on the page, like a real notebook.

Key capabilities:

- Organized in Notebooks, Sections, and Pages — mimicking a physical ring binder.
- Shared notebooks — a whole team can contribute to and access a shared OneNote notebook.
- Integration with Teams — OneNote notebooks are embedded directly into Teams channels as a tab.
- Audio and Video recording — capture meeting audio and sync notes to the recording.
- Web Clipper — save web pages, articles, and screenshots directly into OneNote.

Real-Time Collaboration Across Office Apps

One of the most significant advantages of storing Office files in SharePoint or OneDrive is real-time co-authoring. Multiple users can open and edit the same Word document, Excel spreadsheet, or PowerPoint presentation at the same time. Each person's cursor and edits are visible to all others in real time.

This eliminates the old workflow of emailing files back and forth, merging changes manually, or dealing with conflicting versions. Everyone always works on the same, most up-to-date file.

> **Service Desk Context:** Users occasionally ask why they cannot edit a file — the common causes are: the file is open in a desktop app by another user who has not enabled co-authoring, the file is stored locally (not in SharePoint or OneDrive), or the file is in a format that does not support co-authoring (like older .doc format). Always check where the file is stored when troubleshooting co-authoring issues.

Topic 10: Security and Compliance [Good to Know]

Security and Compliance in Microsoft 365 is a broad area managed primarily by security and compliance teams — it is generally outside the day-to-day scope of Service Desk. However, having a foundational understanding is important because you will encounter these concepts when supporting users, responding to security incidents, or escalating tickets to the right team.

> **Note:** This topic is covered at an awareness level. The goal is to understand what these features are, why they exist, and when they are relevant — not to configure or manage them.

The Microsoft 365 Security Framework

Microsoft 365 takes a layered approach to security. Rather than relying on a single security measure, multiple layers work together to protect identities, data, devices, and applications:

- Identity Protection — Ensuring only the right people get in. MFA, Conditional Access, and sign-in risk policies protect against unauthorized access.
- Threat Defense — Blocking malicious content before it reaches users. Defender for Office 365 scans emails, attachments, and links for threats.
- Data Security — Protecting data wherever it lives. Encryption at rest and in transit, DLP policies, and sensitivity labels prevent data leakage.
- Compliance — Meeting legal and regulatory requirements. Retention policies, eDiscovery, and audit logs support organizations in regulated industries.

Microsoft Defender for Office 365

Microsoft Defender for Office 365 is a security service that protects users against email-based threats — which are the most common attack vector in modern organizations. It works transparently in the background, analyzing every email before it reaches the user's inbox.

Defender Feature	What It Does
Safe Attachments	Every email attachment is opened in a virtual sandbox environment before being delivered. If the attachment is malicious, it is blocked. The user never sees the dangerous file.
Safe Links	Every URL in an email is scanned at the time the user clicks it. If the link has become malicious (even after delivery), Defender blocks the page and warns the user.
Anti-Phishing	Detects and blocks emails that impersonate real people or brands (CEO fraud, fake invoice emails, domain spoofing) — protecting users from social engineering attacks.

Defender Feature	What It Does
Anti-Spam	Filters bulk and unsolicited emails before they reach inboxes. Part of Exchange Online Protection (EOP), which is included in all Microsoft 365 plans.
Anti-Malware	Scans all email content for known malware signatures and blocks infected messages from being delivered.

Data Loss Prevention (DLP)

Data Loss Prevention (DLP) policies are rules that prevent sensitive organizational data from being accidentally or intentionally shared outside the organization. DLP scans content in emails, SharePoint files, OneDrive documents, and Teams messages — and takes action when it detects sensitive information.

Examples of what DLP can detect and protect:

- Credit card numbers or bank account details in an email attachment.
- National ID numbers or passport numbers in a shared SharePoint document.
- Medical or health information in a Teams message.
- Confidential financial reports being shared with external recipients.

When DLP detects a potential violation, it can:

- Warn the user with a policy tip — 'This email appears to contain sensitive information. Are you sure you want to send it?'
- Block the action entirely — prevent the email from being sent or the file from being shared.
- Generate an alert for the security team to review.

> **Example:** An HR employee accidentally attaches a spreadsheet containing employee salary information to an external email. The DLP policy detects the sensitive data and blocks the email from being sent, showing the user a policy tip explaining why.

Retention Policies

A Retention Policy is a rule that tells Microsoft 365 how long to keep data and what to do with it after that period expires. Retention policies apply across Exchange mailboxes, SharePoint sites, OneDrive, and Teams conversations.

Retention policies serve two primary purposes:

- Compliance and Record-Keeping — Many industries and regulations require organizations to keep certain data for a minimum period (e.g., financial records for 7 years, healthcare records for longer). Retention policies enforce this automatically.
- Data Cleanup — After the retention period ends, data can be automatically deleted to reduce storage costs and minimize security risk from old, unused data.

A retention policy can be configured to:

- Keep content for a fixed period — even if a user deletes it.
- Delete content automatically after a certain period.
- First retain, then delete — keep for 7 years, then permanently delete.

> **Example:** A healthcare organization must retain all patient-related communication for 7 years as per regulatory requirements. The compliance team creates a retention policy that applies to all Exchange mailboxes and SharePoint sites. Even if a user deletes an email, it is silently preserved for 7 years before being permanently removed.

Litigation Hold

Litigation Hold is a compliance feature that prevents specific mailbox content from being permanently deleted — typically when an organization is involved in a legal case, investigation, or audit. It is applied to individual mailboxes, unlike retention policies which can be applied broadly.

How Litigation Hold works:

- An administrator places a specific mailbox on Litigation Hold.
- From that point, anything the user deletes or modifies is preserved in a hidden, protected area of the mailbox called the Recoverable Items folder.
- The user continues working normally — they do not notice any difference.
- The compliance or legal team can search and retrieve those preserved items when needed.
- Litigation Hold can be set for a specific duration or kept indefinitely.

Litigation Hold vs Retention Policy: These two are often confused. Litigation Hold is typically applied to specific mailboxes for a defined legal purpose — it is temporary and reactive (triggered by a legal event). Retention Policies are proactive, organization-wide rules applied consistently across all users for compliance and governance purposes.

Example: A company receives a legal notice about a contract dispute involving a project manager. The IT team places the project manager's mailbox on Litigation Hold for 12 months. Even if the manager clears their deleted items, all original messages are preserved. When the legal team needs evidence, the emails are recoverable.

eDiscovery

eDiscovery (Electronic Discovery) is the process of searching, identifying, and exporting electronic data for use in legal proceedings or compliance investigations. Microsoft 365 provides eDiscovery tools through Microsoft Purview that allow authorized personnel to search across mailboxes, Teams conversations, SharePoint sites, and OneDrive.

eDiscovery allows:

- Searching across all Microsoft 365 data sources simultaneously.
- Applying legal holds to prevent data from being modified or deleted during an investigation.
- Exporting search results in formats suitable for legal review.

Microsoft 365 Compliance Center — Microsoft Purview

Microsoft Purview (previously called the Microsoft 365 Compliance Center) is the central portal for all compliance-related settings and activities. Security and compliance administrators use Purview to configure DLP policies, retention labels, eDiscovery cases, audit logs, and data governance settings.

Service Desk Awareness: As a Service Desk engineer, you will not configure DLP or retention policies yourself — these are managed by the compliance team or senior administrators. However, you should know these features exist because you may encounter tickets such as: 'A user says they cannot send an email with an attachment' (may be a DLP policy), 'A user's deleted emails are still appearing in recoverable items' (may be a Litigation Hold or retention policy). Knowing these concepts helps you correctly categorize and escalate the ticket.

Topic 11: Microsoft Intune — MDM and MAM

As organizations move to cloud-based work and employees access company data from personal laptops, smartphones, and tablets — managing and securing those devices has become a critical IT function. Microsoft Intune is the cloud-based service in Microsoft 365 that handles this.

Intune allows IT teams to manage devices and applications centrally — enforcing security policies, ensuring compliance, and protecting company data — regardless of whether the device is a company-owned laptop or an employee's personal phone.

Simple Definition: Think of Intune as the tool that gives IT control over the devices and apps that touch company data. Without Intune, a user's phone is completely outside IT's control — if it is lost or stolen, company data on it has no protection. With Intune, IT can enforce security standards and remotely wipe company data if needed.

What is Microsoft Intune?

Microsoft Intune is a cloud-based Unified Endpoint Management (UEM) solution. It enables organizations to manage and secure a wide range of devices — Windows PCs, Mac computers, iOS iPhones and iPads, and Android smartphones and tablets — all from a single, centralized admin portal.

Intune supports two main management approaches:

- Mobile Device Management (MDM) — Managing and securing the entire device.
- Mobile Application Management (MAM) — Managing and securing only the corporate applications and data on a device, without touching the rest of it.

These two approaches can be used together or independently, depending on whether the device is company-owned or personally owned.

Mobile Device Management (MDM)

MDM is about enrolling the entire device into Intune so that IT has visibility and control over it. Once a device is enrolled, the IT team can push policies and settings to it, monitor its compliance status, and take remote actions on it.

What MDM enables IT to do:

- Enforce security requirements — require a PIN, password, or biometric lock. Require device encryption.
- Push configuration settings — automatically configure Wi-Fi profiles, VPN settings, email accounts, and certificates without the user having to do anything manually.
- Deploy and manage apps — remotely install approved business applications to enrolled devices. Block or remove unauthorized apps.
- Compliance monitoring — continuously check whether a device meets organizational security standards (e.g., is the OS up to date? Is the device encrypted? Is it jail-broken or rooted?).
- Remote actions — lock a device, reset a PIN, or completely wipe the device if it is lost or stolen.

When MDM is the right approach:

- For company-owned devices where IT needs full control.
- When deploying standardized configurations across a fleet of devices.
- When strict compliance requirements must be enforced (e.g., all company laptops must have encryption enabled and an updated OS).

> **Example:** A logistics company provides company-owned tablets to its delivery drivers. All tablets are enrolled in Intune MDM. IT remotely installs the delivery-route app, enforces a PIN lock, blocks access to social media apps, and can wipe the tablet completely if it is reported lost.

Mobile Application Management (MAM)

MAM is a lighter-touch approach that protects corporate data inside specific applications — without enrolling or controlling the entire device. This is particularly important in Bring Your Own Device (BYOD) scenarios, where employees use their personal phones or laptops for work.

With MAM, the IT team does not touch the personal device itself — they only apply protection policies to the Microsoft apps installed on it (such as Outlook, Teams, and OneDrive).

What MAM enables IT to do through app protection policies:

- Require a PIN or biometric to open a managed app — e.g., the user must enter a PIN to open the Outlook app on their personal phone.
- Prevent copy-pasting between managed apps and personal apps — corporate email text cannot be copied into WhatsApp or a personal notes app.
- Block saving company files to personal storage — files from OneDrive cannot be saved to personal Google Drive or the device's camera roll.
- Selective Wipe — if an employee leaves the company or the device is lost, IT can remove only the corporate data from the managed apps, leaving personal photos, contacts, and messages completely untouched.

When MAM is the right approach:

- For personal (BYOD) devices where IT cannot enroll the full device.
- When the organization needs to protect corporate data without invading employee privacy.
- When employees use personal phones to check work email or access Teams.

> **Example:** A salesperson uses their personal iPhone to access work email via Outlook and collaborate on Teams. The company applies MAM policies. The salesperson must enter a PIN to open Outlook, cannot paste email content into personal apps, and cannot save company files to personal storage. When the salesperson leaves the company, IT performs a selective wipe — removing only Outlook and Teams data — while all personal photos, contacts, and apps remain completely intact.

MDM vs MAM — Key Differences

Aspect	MDM	MAM	When to Use
What is managed	The entire device	Only the apps and their data	MDM = company-owned devices. MAM = personal devices.
Device enrollment	Required — device must be enrolled	Not required — no enrollment needed	MAM is better for BYOD to respect privacy.

Aspect	MDM	MAM	When to Use
User privacy	IT can see device details	IT only sees app activity, not the device	MAM preserves personal privacy.
Data wipe capability	Full device wipe — everything gone	Selective wipe — company data only	MAM protects user's personal content.
Best for	Company-owned laptops, phones	Employee's personal phones, BYOD scenarios	Many orgs use both together.

Common Together: Many organizations use MDM and MAM together. Company-owned devices are enrolled in MDM for full control. Personal devices used for work are managed with MAM policies only — protecting corporate data without controlling the user's personal device.

Topic 12: Hybrid Identity

Most large organizations do not operate in a purely cloud-based environment. They have existing on-premises infrastructure — including on-premises Active Directory — that cannot simply be switched off overnight. Hybrid Identity is the approach that connects this on-premises environment with Microsoft 365 in the cloud, giving users a single, unified identity that works across both.

Simple Definition: Hybrid Identity means a user has one username and one password — and it works whether they are logging into their office computer on-premises or accessing Microsoft 365 from home. The identity exists in both places simultaneously, kept in sync automatically.

Why Hybrid Identity is Needed

Without Hybrid Identity, an organization with both on-premises AD and Microsoft 365 would face a difficult situation: users would have two separate accounts — one in on-premises Active Directory and one in Microsoft Entra ID (the cloud directory). They would need two different passwords, and IT would have to manage both separately.

Hybrid Identity solves this by synchronizing the on-premises Active Directory with Microsoft Entra ID, so that:

- Users have a single identity that works everywhere — same username, same password.
- Admins only manage identities in one place — changes propagate to the cloud automatically.
- Users get a seamless experience — they log in once and access both on-premises resources and cloud services without being prompted to authenticate again.

Directory Synchronization — How it Works

Synchronization is the process of automatically copying user account information from on-premises Active Directory to Microsoft Entra ID. A synchronization agent installed on an on-premises Windows Server performs this process — reading users, groups, and attributes from AD and securely sending them to Entra ID.

What gets synchronized:

- User accounts — names, email addresses, job titles, departments.
- Group memberships — security groups and distribution lists.
- Password information — depending on the authentication method chosen.

The synchronization runs on a regular schedule (typically every 30 minutes) and is outbound — the sync server connects out to Microsoft 365. No inbound connections from the internet to your internal servers are required, which is a significant security benefit.

Entra Connect Sync

Entra Connect Sync — previously known as Azure AD Connect — is the traditional, full-featured synchronization tool that has been the industry standard for hybrid identity for many years. It is a Windows Server application that is downloaded from Microsoft's website and installed on a dedicated on-premises server.

Once installed, Entra Connect Sync runs a full synchronization engine locally — meaning all the logic for reading, processing, and mapping identity data happens on the on-premises server before it is sent to Microsoft Entra ID.

How the sync works with Entra Connect Sync:

The process follows a clear sequence every time a synchronization cycle runs — by default, this happens every 30 minutes.

Step 1 — Import: The sync engine connects to the on-premises Active Directory and reads all user objects, group objects, and their attributes. This is called reading from the "connector space" — a local staging database where the raw AD data is held temporarily.

Step 2 — Synchronization Rules: The sync engine applies a set of rules — called synchronization rules — that determine which objects to sync, which attributes to include, and how to map on-premises attributes to their cloud equivalents. For example, the on-premises "displayName" attribute maps to the "displayName" field in Entra ID.

Step 3 — Filtering: Not everything in AD needs to go to the cloud. Filtering rules are applied to exclude certain OUs (Organizational Units), specific groups, or test accounts that should not be synchronized.

Step 4 — Metaverse: After applying rules and filters, the engine builds what is called the Metaverse — a consolidated, normalized view of all objects that are ready to be synchronized. Think of the Metaverse as a staging area where the final version of each identity is assembled before being sent to the cloud.

Step 5 — Export: The processed identity data is exported from the Metaverse and pushed securely to Microsoft Entra ID over an outbound HTTPS connection. Entra ID then creates or updates the corresponding cloud user accounts.

Key characteristics of Entra Connect Sync:

- The entire sync engine runs on the on-premises server — full processing power and control is local.
- Supports complex customization — organizations can write custom synchronization rules and attribute mappings for advanced scenarios.
- Supports multiple on-premises AD forests being synchronized into a single Entra ID tenant.
- Requires a well-maintained, dedicated Windows Server to run reliably.
- Has been widely deployed in large enterprises for many years and is a proven, stable solution.
- Best suited for organizations with complex identity requirements, legacy configurations, or advanced attribute customization needs.

Entra Cloud Sync

Entra Cloud Sync is the modern, next-generation alternative to Entra Connect Sync. Microsoft developed it to address the growing need for a simpler, lighter, and more flexible synchronization tool — particularly for organizations that want to reduce their on-premises infrastructure footprint.

The fundamental difference from Entra Connect Sync is where the processing happens. In Entra Cloud Sync, the heavy lifting — the synchronization logic, attribute mapping rules, and configuration — is handled by Microsoft's cloud service, not by the on-premises server. The on-premises component is just a lightweight agent whose only job is to read data from AD and pass it to the cloud.

How the sync works with Entra Cloud Sync:

Step 1 — Cloud Service Initiates: The Microsoft Entra cloud service determines what needs to be synchronized based on the configuration set up in the Entra admin portal. The cloud service is in charge — it tells the on-premises agent what to do.

Step 2 — Agent Reads from AD: The lightweight Cloud Sync agent, installed on an on-premises Windows Server, receives the instruction from the cloud service and reads the relevant user objects, groups, and attributes from the on-premises Active Directory.

Step 3 — Data Sent to the Cloud: The agent securely transmits the raw identity data to the Microsoft Entra cloud service over an outbound HTTPS connection. No inbound ports need to be opened on the firewall.

Step 4 — Cloud Applies the Logic: The Entra cloud service applies attribute mapping rules, filters, and transformation logic in the cloud. This is the key difference — all of the processing that happens on-premises in Entra Connect Sync happens in the cloud with Entra Cloud Sync.

Step 5 — Entra ID is Updated: The cloud service updates Microsoft Entra ID with the final, processed identity data — creating or updating user accounts as needed.

Key characteristics of Entra Cloud Sync:

- The on-premises agent is lightweight — it requires minimal server resources and is simple to install and maintain.
- Configuration is done entirely through the Microsoft Entra admin portal in the cloud — no local configuration files or complex tools needed.
- Supports multiple on-premises AD forests syncing to a single Entra ID tenant — useful for organizations that have grown through mergers and acquisitions.
- Easier to deploy in distributed environments — multiple lightweight agents can be installed across different sites for redundancy.
- Less customization compared to Entra Connect Sync — advanced attribute mapping scenarios may not be supported.
- Microsoft's recommended direction for new deployments and simpler environments going forward.

Side-by-Side Comparison

Aspect	Entra Connect Sync	Entra Cloud Sync
Where sync logic runs	On-premises server	Microsoft's cloud service
On-premises component	Full sync engine (heavy)	Lightweight agent only
Configuration	Local tools and config files	Entra admin portal (cloud-based)
Customization	High — advanced rules supported	Limited — standard scenarios
Multi-forest support	Yes	Yes
Server requirement	Dedicated, well-maintained server	Minimal — any Windows Server
Best for	Complex, large enterprises	Simpler environments, new deployments
Microsoft's direction	Established, still supported	Recommended for new setups

Authentication Methods in Hybrid Identity

In a hybrid environment, an important decision is where authentication actually happens — does the user's password get validated on-premises or in the cloud? Microsoft provides three options:

1. Password Hash Synchronization (PHS) — Most Common

In PHS, the on-premises Active Directory password is processed into a secure hash, and that hash is synchronized to Microsoft Entra ID. When users sign in to Microsoft 365, their credentials are validated directly against Entra ID in the cloud.

- The actual password is never sent to the cloud — only a secure, non-reversible hash.
- Authentication happens entirely in the cloud — fast and reliable.
- Works even if on-premises servers are temporarily unavailable (cloud-side sign-ins are not affected).
- Simplest to implement and maintain — Microsoft's recommended option for most organizations.

> **Why Some Organizations Avoid PHS:** Even though no actual password is sent to the cloud, some regulated organizations (government agencies, defense contractors, certain financial institutions) have strict policies that prohibit any password-related value from leaving their network. These organizations choose PTA or Federation instead.

2. Pass-Through Authentication (PTA)

In PTA, when a user signs in to Microsoft 365, the sign-in request is sent through Entra ID — but then forwarded to a lightweight authentication agent installed on-premises. That agent validates the password directly against the on-premises Active Directory, and the result is returned to Entra ID.

- No password or password hash is ever stored in the cloud.
- Password validation always happens on-premises — subject to on-prem policies like login hours or account restrictions.
- If on-premises servers are unavailable, authentication may fail for some users.
- Preferred by organizations that must ensure passwords are validated entirely within their own network.

3. Federation with AD FS

In a federated setup, Microsoft 365 delegates all authentication responsibility to an on-premises Active Directory Federation Services (AD FS) server. When users sign in, they are redirected to the AD FS server, which validates their credentials and issues a security token back to Microsoft 365.

- No password or hash is stored in Entra ID — authentication is entirely on-premises.
- Most complex setup — requires dedicated AD FS servers and additional infrastructure.
- Provides maximum control over authentication policies and flows.

- Still found in large, security-sensitive enterprises — though PHS and PTA have improved enough that many organizations are moving away from federation.

Method	Where Authentication Happens	Key Consideration
Password Hash Sync (PHS)	Cloud — Microsoft Entra ID	Simplest, most reliable, works offline. Recommended for most organizations.
Pass-Through Auth (PTA)	On-Premises — AD validation agent	No cloud password storage. Requires on-prem infrastructure available during sign-in.
Federation — AD FS	On-Premises — AD FS server	Maximum control. Most complex. High infrastructure overhead.

Service Desk Context: Understanding Hybrid Identity is important for troubleshooting. If a user can sign in to on-premises systems (their laptop, internal applications) but cannot access Microsoft 365, the issue may be related to synchronization (their account has not synced to Entra ID) or authentication (a PTA agent is down, or there is an AD FS issue). Conversely, if they can access Microsoft 365 but not on-prem systems, the issue is likely with their local AD account.